

Physiology: Urine Concentration & Dilution

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Session Objectives

At the conclusion of the session, the student should be able:

- 1. Define & describe the origin of antidiuretic hormone and the major controls of its secretion**
- 2. Define & Describe how antidiuretic hormone affects water and urea reabsorption.**
- 3. Describe the characteristics of the medullary osmotic gradient.**
- 4. Explain the role of the thick ascending limb, urea recycling, and medullary blood flow in generating the medullary osmotic gradient.**
- 5. State why the medullary osmotic gradient is partially “washed out” during water diuresis.**

You've just spent your last "free" summer trying to get back into shape, as your diet and exercise habits in your 1st year of medical school were horrible. It has been a very hot summer with multiple heat waves.

One very hot afternoon (Heat Index of 103.5°), you decide to go jogging after a session practicing hot yoga. When you leave the studio, you are light-headed and you feel your heart "skipping beats".

Your yoga instructors give you a couple of bananas and a bottle of coconut water. Your urine output is dark yellow, almost orange. After about 5', you feel better but you're still extreme thirsty. You get home and start drinking cold water. You drink around 2 gallons of water before you start to urinate.

Your urine get less dark as the goes on until it's light yellow. You start thinking on how and why this occurred.







H₂O Balance and Renal Handling of H₂O

- In the steady state, H₂O intake and output must be equal.
- The body's three major sources of H₂O are:
 - Ingested H₂O_L.
 - H₂O contained in the food.
 - H₂O produced by aerobic metabolism.
- Major route of H₂O loss: Kidneys.
- Minor route of H₂O loss: Feces, exhaled air
- Sweat production:
 - Geared to help regulate body core T°, not H₂O balance.
 - Production of sweat can increase markedly during exercise or at high temperatures
 - These amounts vary, depending on diet, physical activity, and the environment (e.g., temperature and humidity).
- Renal H₂O output can compensate for high or low H₂O intake
- Kidneys excrete a variable amount of solutes, depending on intake.

INPUT	
Source	Amount (ml)
Ingested fluids	1200
Ingested food	1000
Metabolism	300
Total	2500
OUTPUT	
Route	Amount (mL)
Urine	1500
Feces	100
Skin/sweat	550
Exhaled air	350
Total	2500

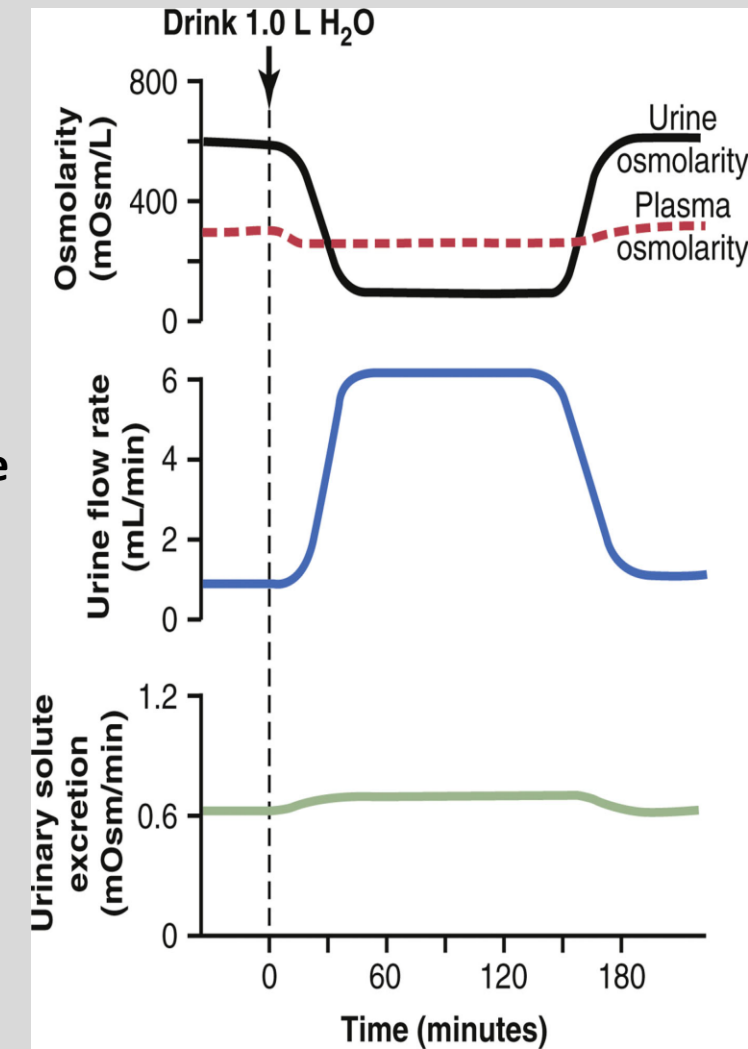
H₂O Balance and Renal Handling of H₂O

Example

- “Normal” diet: Excreted solute ~ 600 mosm/day.
- Excreted solutes are dissolved in a daily urine output (~1500 mL).
- Key principle:
 - Regardless of the volume of excreted H₂O, kidneys must excrete ~600 mosm/day.
 - ∴ to excrete a wide range of H₂O volumes, kidneys must produce urine having a wide range of osmolalities.
- When the intake of H₂O is high → kidneys generate urine w/ low osmolality (~ 40 mOsm).
- To conserve H₂O (e.g., restricted H₂O intake or excessive loss), kidneys generate urine with an osmolality as high as ~1200 mOsm (w/ average solute load)
- Minimal urine volume ~0.5 L/day.
- Renal failure ∴ reduces both concentrating & diluting ability.

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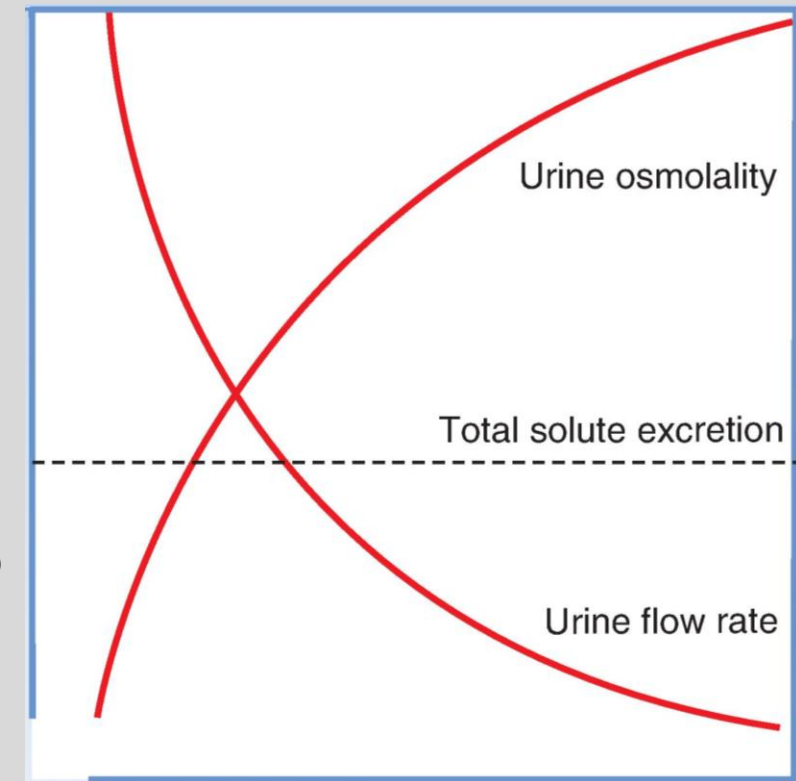


Free H₂O Clearance

- **Sensible H₂O loss:**
 - H₂O loss through kidneys, GI tract & sweat glands (person is aware of its occurrence).
 - Tightly regulated process.
- **Insensible H₂O loss:**
 - H₂O loss via evaporation from the skin and exhaled thru respiration (individual is unaware of its occurrence)
 - Increases in hot environments, exercise, or in the presence of fever.
 - Cannot be regulated.
- **Urine consists of two moieties:**
 - Volume necessary to dissolve all excreted solutes at a concentration that is isosmotic with blood plasma (osmolal clearance; C_{osm})
 - Volume of pure (solute-free H₂O, free H₂O) that one must add (or subtract) to the previous volume to account for the entire volume (free H₂O clearance; (C_{H_2O}))

$$\text{Urine Output (V)} = C_{Osm} + C_{H_2O}$$

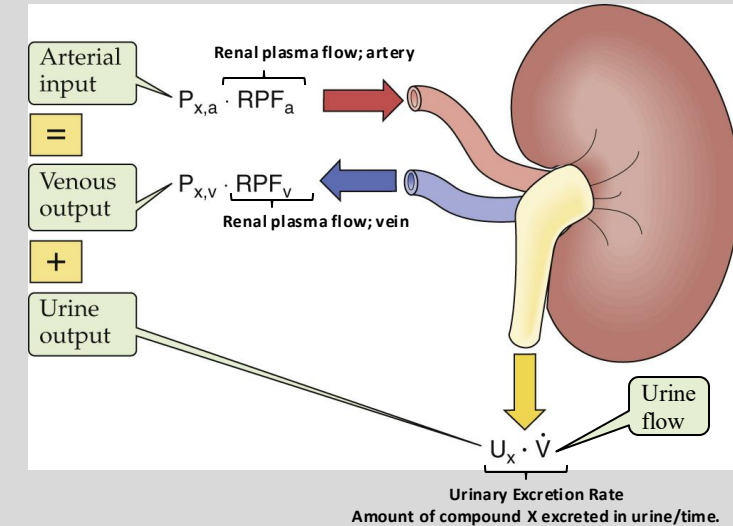
- The clearance of any compound is the virtual volume of plasma that would be totally cleared of that compound in a unit of time (e.g., ml/min, L/hr...)



Free H₂O Clearance

- C_{H_2O} is positive:
 - Kidneys produce urine that is less concentrated than plasma ($< 300 \text{ mOsm}$)
- C_{H_2O} is negative:
 - Kidneys produce urine that is more concentrated than plasma ($> 300 \text{ mOsm}$)
- The 3 basic functions of the kidney (GFR, tubule reabsorption & tubule secretion) determine the renal clearance of a compound.
- C_{H_2O} is defined as the clearance of H₂O devoid of all solutes.
- The range of C_{H_2O} values for kidneys is related to urine osmolality:
 - Isosmotic Urine:
 - If the osmolalities of the urine and plasma are the same ($U_{Osm} = P_{Osm}$), then $C_{Osm} = V$ ($C_{H_2O} = 0$)
 - Dilute Urine:
 - If the urine is more dilute than plasma ($U_{Osm} < P_{Osm}$), the difference between urine flow (V) and C_{Osm} is positive.
 - Concentrated Urine:
 - If the urine is more concentrated than plasma ($U_{Osm} > P_{Osm}$), then the difference between urine flow (V) and C_{Osm} is negative.

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$$\dot{V} = C_{Osm} + C_{H_2O}$$

$$C_{H_2O} = \dot{V} - C_{Osm}$$

If we compute C_{Osm} the same way we would compute the clearance of any substance

$$C_{Osm} = \frac{U_{Osm} \cdot \dot{V}}{P_{Osm}}$$

BUT

$$C_{H_2O} \neq (U_{H_2O} \cdot \dot{V}) / P_{H_2O}$$

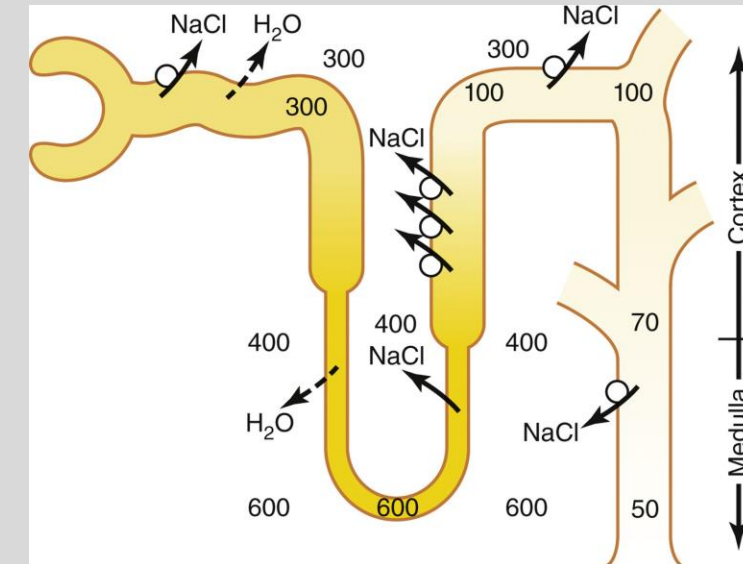
We can obtain the C_{H_2O} only by subtraction

H₂O Transport in Nephron Segments

- Generation of dilute urine:
 - Pumping of solutes out of the tubular lumen in nephron segments that are impermeable to H₂O
 - Tubular fluid is hypo-osmotic vs blood.
- Generation of concentrated urine:
 - Driving H₂O (via osmosis) from the tubular lumen into a hyperosmotic interstitium.

Concentrated urine:

- Pump H₂O actively out of the tubule lumen? NOPE!! (H₂O pumps do not exist).
- Kidney uses osmosis as the driving force to concentrate tubular lumen contents.
- The creation of an osmotic gradient to generate a hyperosmotic interstitial fluid in a confined compartment (renal medulla).
- Final step for making concentrated (hyperosmotic) urine:
 - Controlled by regulated water permeability
 - Allows luminal content of medullary collecting duct (MCD) to equilibrate with the hyperosmotic interstitium → concentrated urine.



H₂O Transport in Nephron Segments

- Net absorption of H₂O occurs all along the nephron
- However, not all segments alter the osmolality of the tubule fluid.
- The proximal tubule, *regardless of the final osmolality of the urine*, reabsorbs 2/3 of the filtered fluid isosmotically (i.e., the fluid reabsorbed has nearly the same osmolality as plasma).

H₂O Reabsorption $\cong$ Solute Reabsorption

- The loop of Henle and the distal convoluted tubule (DCT) reabsorb salt in excess of H₂O.

H₂O Reabsorption $\ll$ Solute Reabsorption

- Tubular fluid leaving the DCT is hypo-osmotic.
- *Whether the final urine is dilute or concentrated* depends on whether water reabsorption occurs in more distal segments: the initial and cortical collecting tubules (ICT and CCT) and the outer and inner medullary collecting ducts (OMCD and IMCD).

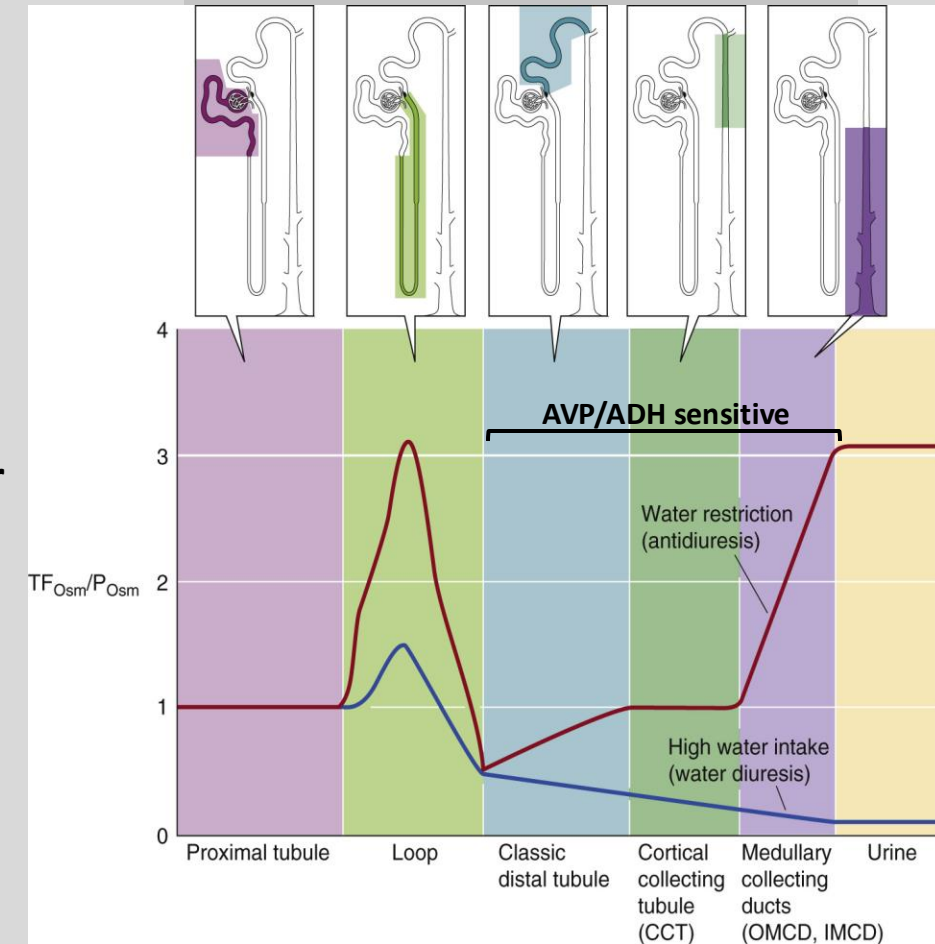
H₂O Reabsorption vs Solute Reabsorption?
(variable)

- Arginine vasopressin (AVP), a.k.a. Antidiuretic hormone (ADH):
 - Regulates the variable fraction of H₂O reabsorption in these 4 segments.

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Δ in Osmolality of TF along the nephron



TF_{Osm} : Osmolality of the tubular fluid

P_{Osm} : Plasma osmolality

H₂O Transport in Nephron Segments

- $TF_{Osm}/P_{Osm} \Delta s$ along most of the nephron in H₂O restriction & excess.

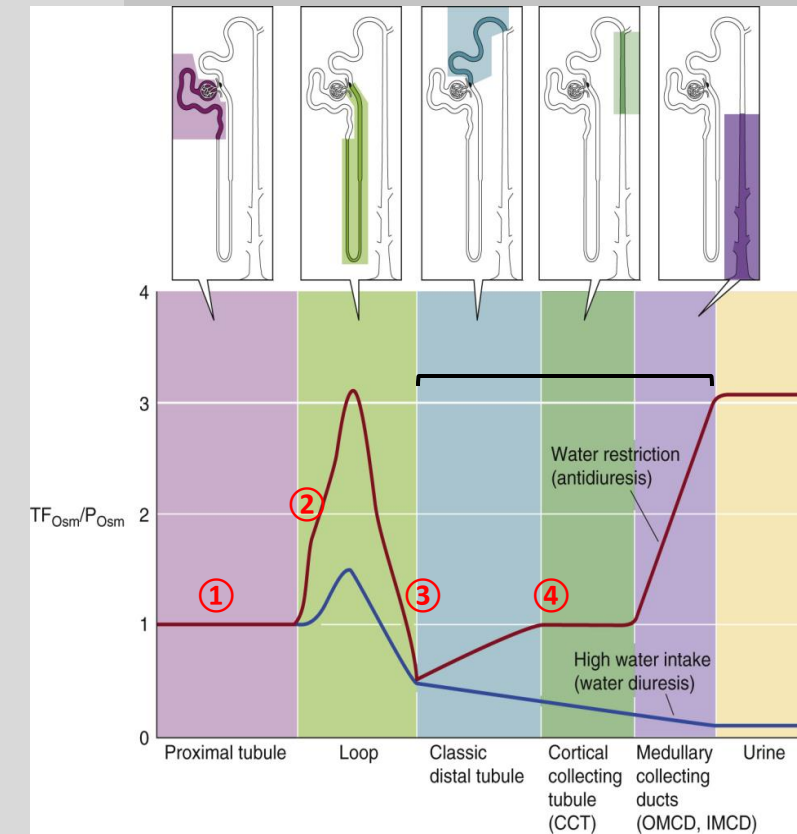
Case of H₂O restriction (hydropenia):

- Kidneys must maximally concentrate urine and excrete a minimal volume of H₂O (antidiuresis)
- Proximal Tubule:
 - Tubular fluid does not change in osmolality ($\leftrightarrow$ concentration) ①
- Thin Descending Limb of Loop of Henle:
 - Tubular fluid becomes hypertonic ($\uparrow$ concentration) ② as the thin limb is *permeable* to H₂O & *impermeable* to electrolytes.
- Thick Ascending Limb of Loop of Henle (diluting segment):
 - Tubular fluid becomes hypotonic ($\downarrow$ concentration) ③ as the thick limb is *impermeable* to H₂O & *permeable* to electrolytes.
- Distal and Collecting Tubules ④
 - Tubular fluid becomes hypertonic ($\uparrow$ concentration) as the dilute fluid in the tALH passes to the distal nephron & distal collecting duct even though there is additional reabsorption of NaCl (via eNAC).
 - Elevated levels of AVP/ADH increase the permeability of H₂O to the sensitive sections of the nephron.
 - Medullary interstitium becomes increasingly more hyperosmotic from the corticomedullary junction to the papillary tip.
 - $\therefore$ tubule fluid osmolality increases

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Δ in Osmolality of TF along the nephron

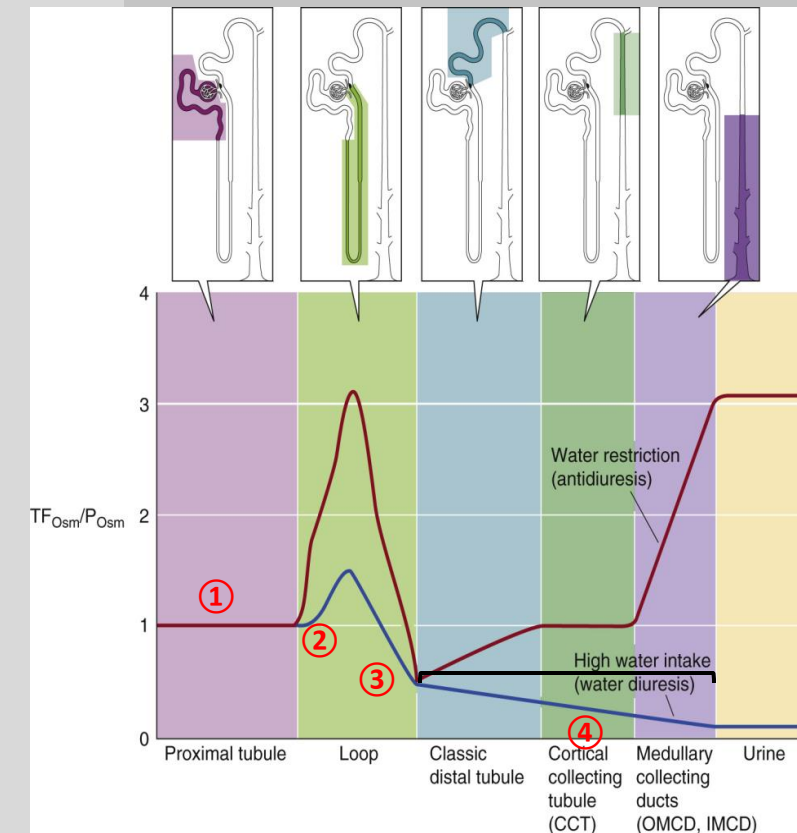


H₂O Transport in Nephron Segments

Case of ingestion of excess H₂O:

- Kidneys must produce a large volume of dilute urine (H₂O diuresis).
- Proximal Tubule:
 - Tubular fluid does not change in osmolality ($\leftrightarrow$ concentration) ①
- Thin Descending Limb of Loop of Henle:
 - Tubular fluid becomes hypertonic ($\uparrow$ concentration) ② as the thin limb is *permeable* to H₂O & *impermeable* to electrolytes.
- Thick Ascending Limb of Loop of Henle (diluting segment):
 - Tubular fluid becomes hypotonic ($\downarrow$ concentration) ③ as the thick limb is *impermeable* to H₂O & *permeable* to electrolytes.
- Distal and Collecting Tubules ④
 - Tubular fluid becomes hypertonic ($\uparrow$ concentration) as the dilute fluid in the tALH passes to the distal nephron & distal collecting duct even though there is additional reabsorption of NaCl (via eNAC).
 - Under conditions of H₂O loading, AVP/ADH levels are depressed
 - This will cause low H₂O permeability of the distal nephron.
 - Continued reabsorption of NaCl along the distal nephron effectively separates salt from H₂O $\rightarrow$ Relatively hypo-osmotic fluid remains in the lumen throughout the remainder of the nephron.

Δ in Osmolality of TF along the nephron

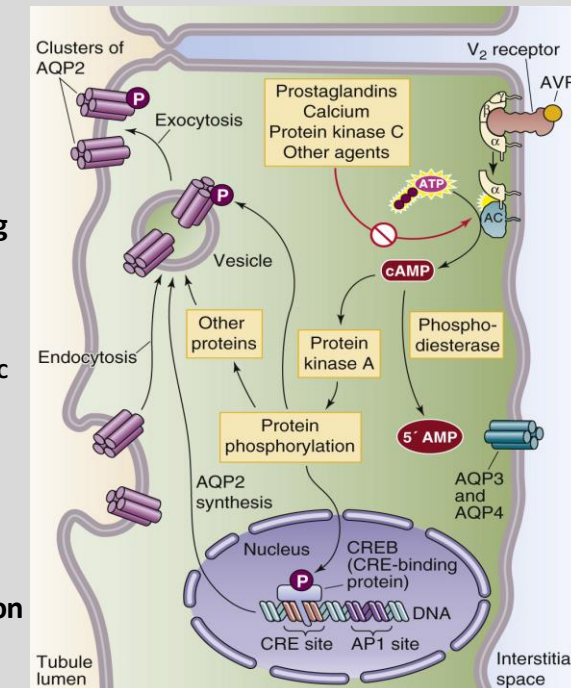
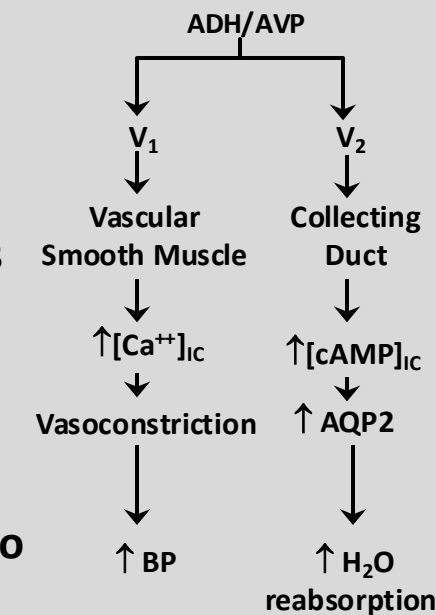
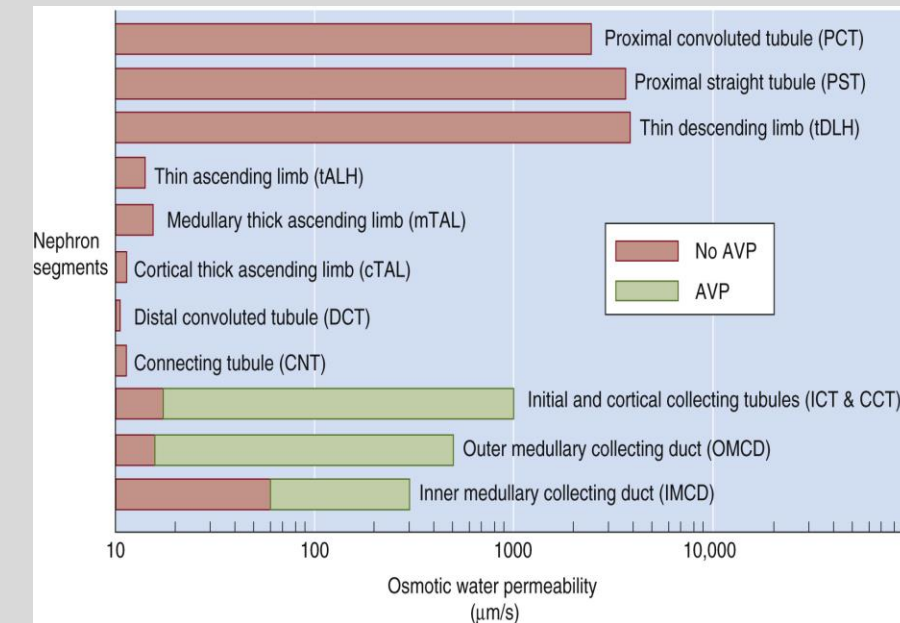


Regulation by Arginine Vasopressin

AVP/ADH:

- Plasma osmolality is tightly regulated.
- Set mainly by the ratio of ECF NaCl to H₂O (glucose and K⁺ also contribute)
- Control of secretion:
 - Neurons responsive to changes in osmolality (osmoreceptors)
 - Located in tissues surrounding the 3rd & 4th cerebral ventricles (circumventricular organs).
 - Circumventricular organs contain fenestrated capillaries, allowing rapid adjustments
- Nonapeptide (9 AA) synthesized in large-bodied neurons in the paraventricular and supraoptic nuclei of the hypothalamus.
- Packaged and transported along the axons to the posterior pituitary.
- Released through the blood-brain barrier into the systemic circulation.
 - High circulating levels (e.g., hypovolemic shock) stimulate V₁ receptors on vascular smooth muscle cells (vasoconstriction).
 - Vasoconstriction → ↑ systemic blood pressure.
 - It also stimulates V₂ receptors on cells of the Collecting Duct.
 - Stimulation will increase cAMP via activation of adenylate cyclase.
 - Inc cAMP → increase transcription and membrane insertion of AQP2 to the basolateral membrane.
 - ↑ H₂O permeability of the collecting tubule → ↑ H₂O reabsorption

H₂O permeability of the nephron

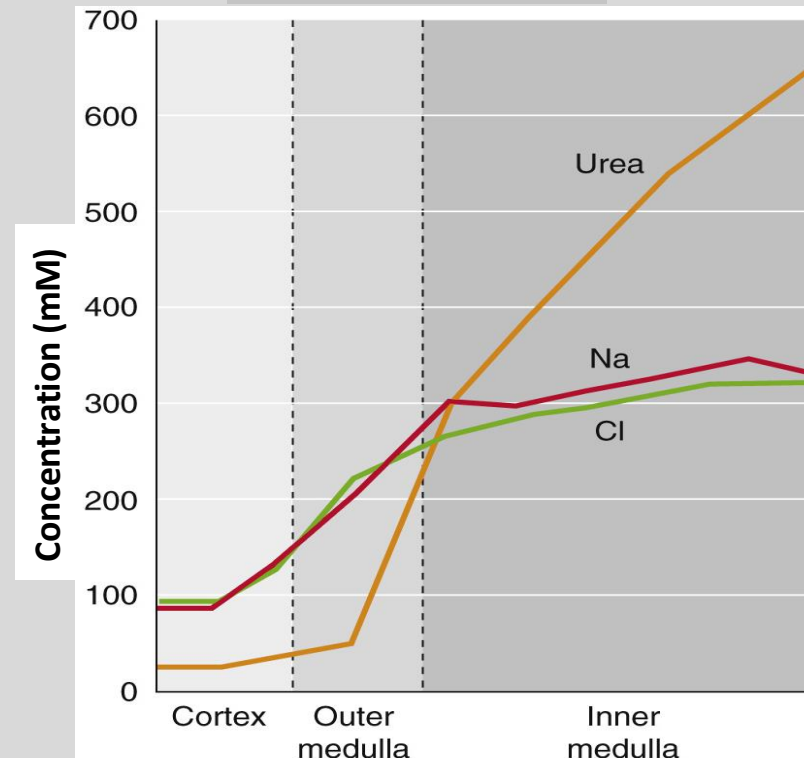


Generation of Hyperosmotic Medulla & Urine

- Understanding the mechanisms involved in forming a hyperosmotic/hypoosmotic urine requires knowing:
 - Solute & H₂O permeability of each tubular segment.
 - Osmotic gradient between the tubule lumen and its surrounding interstitium.
 - Active transport mechanisms that generate the hyperosmotic medullary interstitium.
 - “Exchange” mechanisms that sustain the hyperosmotic medullary compartment.
- Interstitial osmolality progressively rises from the cortex to the tip of the medulla (Corticomedullary/Medullary osmolality gradient).
- The medulla is hyperosmotic to blood plasma during both antidiuresis & diuresis.
- tALH (H₂O impermeable) plays a key role in both dilution & concentration of urine by removal of NaCl.
- Role of Urea:
 - H₂O reabsorption in tDLH → ↑ [Ur]_L → ↑ transepithelial gradient → ↑ Ur reabsorption → ↑ [Ur]_{Interstitium}

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Interstitial Osmolality



Generation of Hyperosmotic Medulla

Main components:

- Active NaCl transport by the thick ascending limb of Henle (tALH).
- Parallel arrangement of blood vessels & tubules.

tALH:

- Reabsorb solute w/o H₂O (adding solute to interstitium).
- IMPERMEABLE to H₂O.
- Dilution of the luminal fluid (urine).

Medullary vasculature:

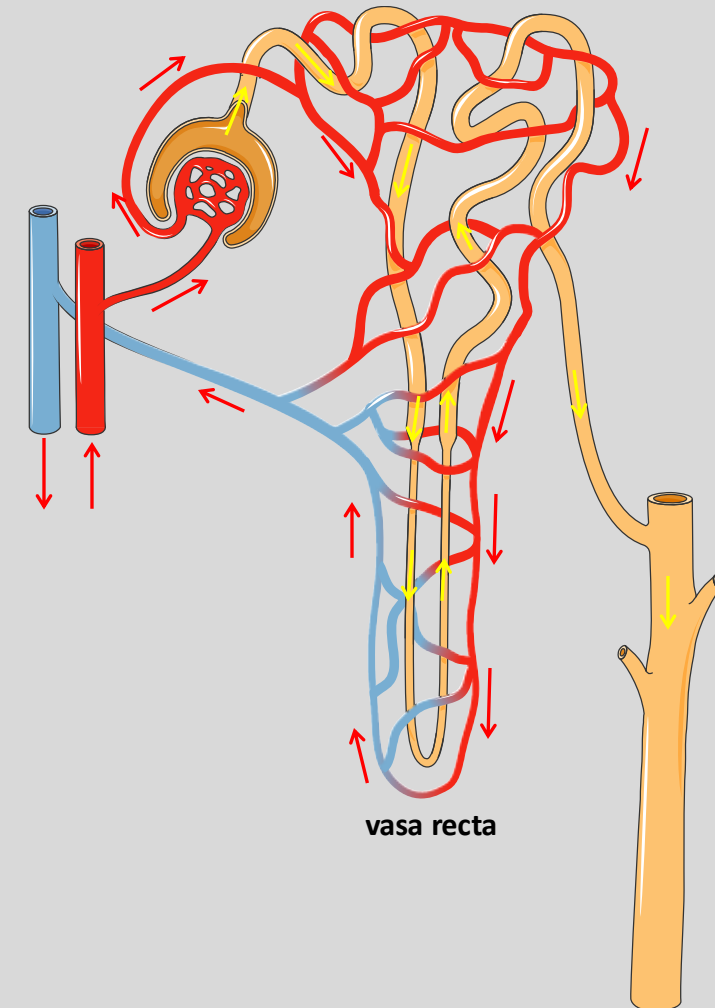
- Arranged differently vs cortex.
- Total blood flow, permeability & volume is lower → accumulation of solutes.

tDLH:

- Freely permeable to H₂O & impermeable to NaCl.
- Cumulative effects result in the countercurrent multiplier system.
- NaCl transport generates only a ~200 mOsm gradient across any portion of the ascending limb
- Countercurrent exchange can multiply this single effect to produce a 900 mOsm gradient between cortex and papilla
- Evolved to create an hyperosmotic urine

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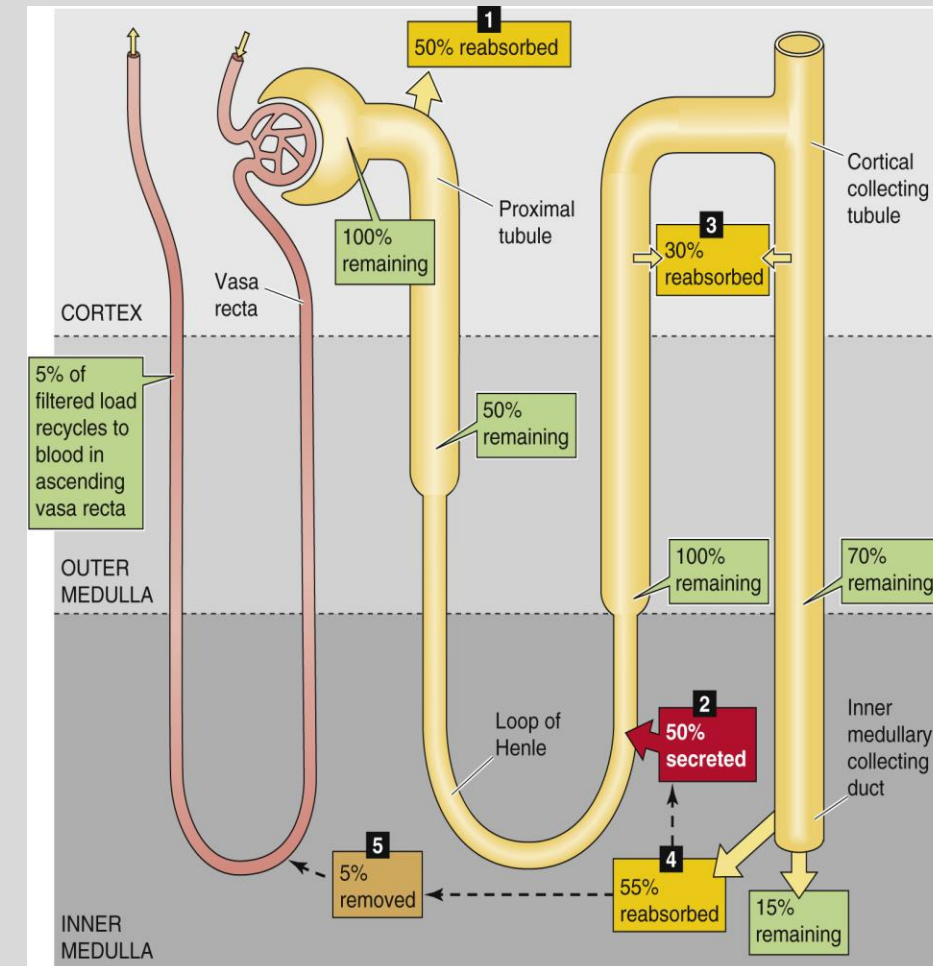


Urea Handling & Hyperosmotic Medulla

- The renal handling of Ur is complex.
- The kidney filters Ur in the glomerulus and reabsorbs ~50% in the PT (1).
- In juxtamedullary nephrons, tDLH & tALH secrete Ur into the lumen (2).
- Some Ur reabsorption occurs along the tALH → CCT (3).
- Finally, the IMCD reabsorbs Ur (4).
- Net Effect: Kidney excretes less Ur into the urine than it filters.
- FE_{Ur} : ~15% (minimal urine flow) → >65% (maximal urine flow).
- During antidiuresis, as tubule fluid enters the tALH, $[Ur]_L$ is several-fold higher than it is in the plasma because ~100% of FL of Ur remains, even though earlier nephron segments have reabsorbed water.
- All nephron segments from the tALH → CD have low permeabilities to Ur.
- In the presence of ADH/AVP, segments from CT → IMCD have high H_2O permeability and continuously reabsorb fluid.
- $\therefore [Ur]_L$ gradually rises, reaching a concentration 8-10X higher than plasma
- ADH/AVP increases H_2O and Ur permeability in the IMCD.
- High $[Ur]_L$ & permeability promote the outward facilitated diffusion of Ur from the IMCD lumen into the medullary interstitium.
- Result: Ur accumulates in the interstitium and contributes about half of the total osmolality in the deepest portion of the inner medulla.

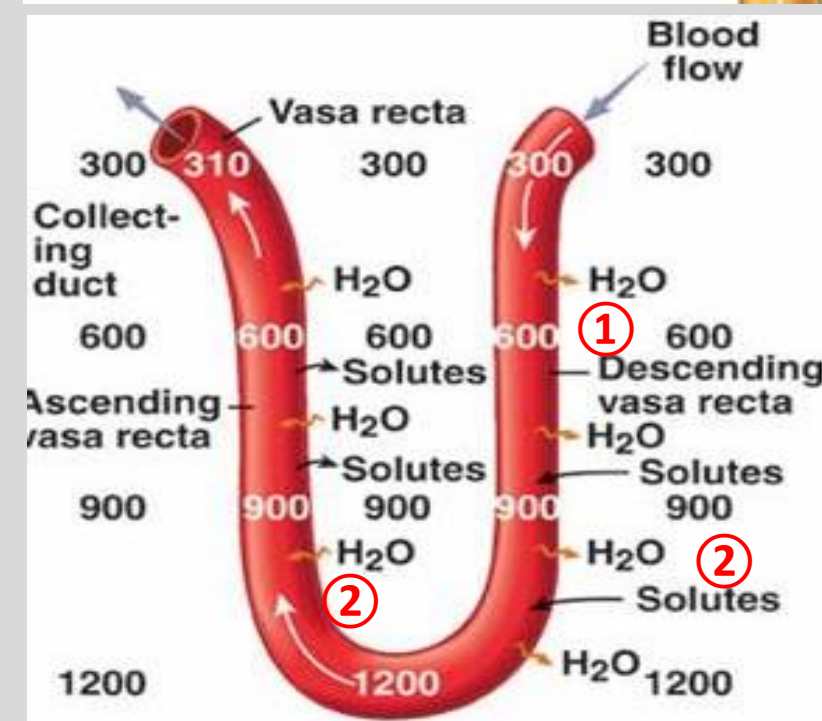
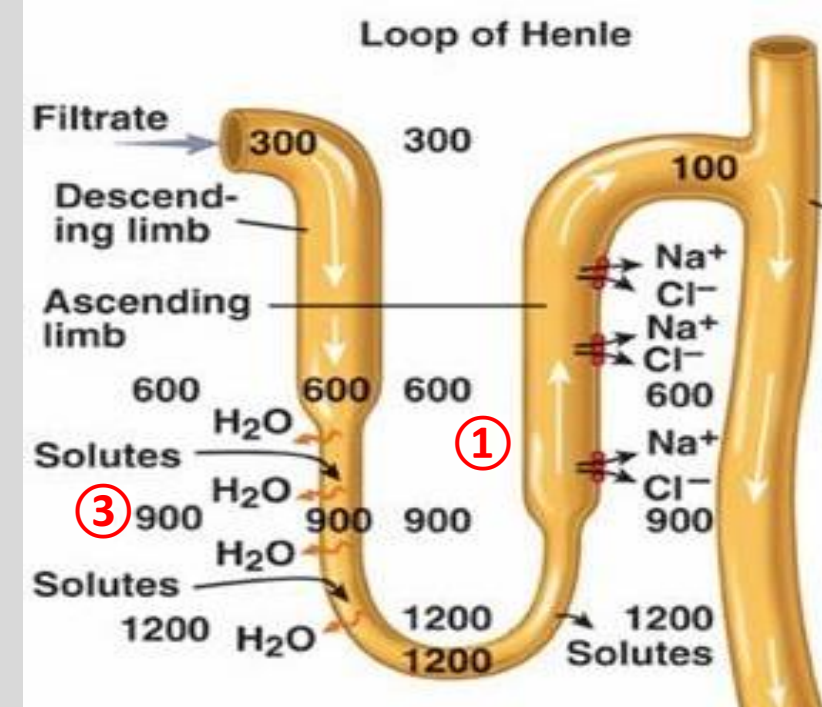
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Countercurrent Multiplier System

- A sequence of events after ultrafiltration at the glomerulus and absorption in the PT.
- Key processes that generate the medullary osmotic gradient:
 - ① Active transport of Na^+ from the loop of Henle into the interstitium of the outer medulla and its distribution deeper into the medulla via descending vasa recta.
 - ② H_2O diffuses from descending to ascending vasa recta (countercurrent movement).
 - ③ Urea recycles from inner medullary collecting ducts to the loop of Henle
- The descending vasa recta carry the salt into the inner medulla and raise the interstitial osmolality everywhere
- As interstitial salt concentration rises, so does salt concentration in the ascending vasa recta.
- Steady state is reached in which salt added to the medulla from the tubules is matched by salt leaving the medulla in ascending vasa recta (taking blood away).
- The salt content of the interstitium is now elevated and constant, and the concentration is considerably higher than in the general circulation.





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